

Hashing

$$Q - 10^0 \times 07 + 10^1 \times 08 + 10^2 \times 09$$

$$10^2 \times 07 + 10^1 \times 08 + 10^0 \times 09$$

$$\text{Load factor } \lambda = \frac{N}{T.S} \rightarrow \frac{\text{total item}}{\text{Table size}}$$

$\lambda > 0.5 \rightarrow$ rehashing in backend
 $\downarrow$
transferring in another table.

open Address

$$h(x) = \left[\text{key} + \underbrace{f(i)}_{\substack{\downarrow \\ \text{conflict} \\ \text{taking help of another.}}} \right] \% \underbrace{T.S}_{\substack{\downarrow \\ \text{table size}}}$$

Key % Size	$\downarrow$ linear probing
$(K+1) \% \text{Size}$	
"	
"	
"	

$n \rightarrow$ data items (total).

Why free space is guaranteed in
linear prob but not in quadratic

OOP 1

separate chaining worst case $O(n+1)$

all are in a single linked
list.

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Rabin Karp

$N \ S_1 \rightarrow abedecc \ adet$

$M \ S_2 \rightarrow bed$

$\text{for } (\underline{s_i})$

$\text{for } (\underline{\quad})$

|

$O(MN)$

↳ instead

$O(M+N) \rightarrow \text{Rabin Karp.}$

$\text{hash}(s_2)$

$\text{hash}(\text{window})$

$s[i] \times 10^0 + s[j] \times 10^1 + s[k] \times 10^2 \dots$

$\dots ? \times 10^9$

Again becomes $O(N \times M)$

$O(10^9 \times 10^9)$ with sum

Rolling hash

$$abcd = a \times 10^0 + \boxed{b \times 10^1 + c \times 10^2 + d \times 10^3}$$

hash
reuse the hash values
divide by base
give

$$bcde = b \times 10^0 + c \times 10^1 + d \times 10^2 + e \times 10^3$$

$$H_i = (H_{i-1} - S_{i-1}) \div b + S_i \times b^{m-1}$$

$$O(\cancel{N} \times N)$$

Next $\rightarrow$ True.